

Phishing Detection & Awareness Report

Future Interns Cybersecurity Internship – Task 2

Project: Phishing Email Detection & Awareness System

Task ID: FUTURE_CS_02

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1. Executive Summary

Phishing is a social-engineering attack in which an attacker attempts to deceive a recipient into revealing sensitive information, opening malicious content, or visiting a fraudulent website.

This assessment analyzes three phishing email examples and identifies the characteristics that can help users recognize phishing attacks.

The analysis focuses on sender identity, social-engineering techniques, suspicious links or attachments, urgency, impersonation, and the potential for credential theft.

The three analyzed examples were classified as phishing based on the available evidence.

2. Objective

The objectives of this assessment are:

- Analyze phishing email examples.
 - Identify common phishing indicators.
 - Classify email risk as Safe, Suspicious, or Phishing.
 - Explain phishing attacks in simple language.
 - Develop practical employee awareness guidance.
 - Recommend safe actions for users who receive suspicious messages.
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3. Scope

The assessment covers:

1. Sender and impersonation indicators.
2. Email subject and message content.
3. Social-engineering techniques.
4. Suspicious URLs.
5. Unexpected attachments.
6. Email-header availability.
7. Risk classification.
8. User prevention and awareness.

The analysis does not intentionally open potentially malicious URLs or attachments.

4. Methodology

The following workflow was used:

Sample Collection
↓
Sample Sanitization
↓
Sender & Content Examination
↓
URL / Attachment Examination
↓
Header Availability Check
↓
Phishing Indicator Identification
↓
Risk Classification
↓
Attack Explanation
↓
Prevention Recommendations

Public phishing examples were used for educational analysis. Third-party content was not claimed as original work.

5. Risk Classification Method

Safe

No significant phishing indicators are identified from the available evidence.

Suspicious

One or more unusual characteristics are present, but available evidence is insufficient for a confident phishing classification.

Phishing

Multiple indicators strongly support a fraudulent or credential-harvesting attempt.

6. Sample 01 – Account Lockout

Source

Educational sample supplied directly in the Future Interns Task 2 description.

Classification

PHISHING – HIGH CONFIDENCE

Indicators

- Urgent account-lockout subject.
- Fear-based language.
- 24-hour deadline.
- Generic greeting.
- Suspicious account-verification domain.
- Request to verify account details through an unexpected link.

Attack Explanation

The message attempts to make the recipient afraid of losing account access. The user is then encouraged to follow a verification link.

A fraudulent destination could potentially request credentials or other sensitive information.

The destination was not opened during this analysis.

User Recommendation

Do not click the link. Open the organization's official website directly or contact the organization's support team through a trusted channel.

7. Sample 02 – Password Expiration

Source

SeanWrightSec/phishing-examples public educational repository.

Classification

PHISHING – HIGH CONFIDENCE

Indicators

- IT-department impersonation.
- Password-expiration theme.
- Urgency.
- Suspicious password-update link.
- Potential credential-harvesting objective.

Attack Explanation

The attacker attempts to convince the recipient that their password needs immediate attention.

The victim may follow the supplied link and enter credentials into a fraudulent login page.

The linked destination was not intentionally opened during this analysis.

User Recommendation

Do not use the email link to update the password. Open the organization's known account portal directly and verify the password status there.

8. Sample 03 – App Store Purchase

Source

SeanWrightSec/phishing-examples public educational repository.

Classification

PHISHING – HIGH CONFIDENCE

Indicators

- Trusted-brand impersonation.
- Unexpected purchase notification.
- Concern about unauthorized financial activity.
- Unexpected PDF attachment.
- Social-engineering pressure.

Attack Explanation

The attacker uses a familiar brand and a supposed purchase to create concern.

The recipient may open the attachment or follow instructions provided by the message.

The attachment was not opened during this educational analysis.

User Recommendation

Do not open unexpected attachments. Verify the transaction directly through the official application or website.

9. Comparative Findings

Indicator	Sample 01	Sample 02	Sample 03
Urgency	Yes	Yes	Indirect
Fear/Concern	Yes	Yes	Yes
Brand/Organization Impersonation	Yes	Yes	Yes
Suspicious Link	Yes	Yes	Not primary
Unexpected Attachment	No	No	Yes
Social Engineering	Yes	Yes	Yes
Credential Theft Risk	High	High	Possible
Classification	Phishing	Phishing	Phishing

10. Common Phishing Indicators

10.1 Unexpected Urgency

Attackers frequently create deadlines to prevent users from thinking carefully.

Examples include:

- Account will be locked.
- Password expires today.
- Payment must be confirmed immediately.
- Security verification required within 24 hours.

Users should slow down and independently verify the request.

10.2 Fear and Threats

Messages may threaten account suspension, financial loss, or security problems.

Users should not allow fear or urgency to replace normal verification procedures.

10.3 Suspicious Sender

Users should check the complete email address instead of trusting only the display name.

A familiar display name does not prove that the message came from the organization.

10.4 Suspicious Links

A link may lead to a domain unrelated to the organization it claims to represent.

Users should avoid opening unexpected links and instead navigate to the official website independently.

10.5 Unexpected Attachments

Unexpected PDFs, documents, archives, or executable files can be used to deliver malicious content or direct users toward fraudulent websites.

10.6 Generic Greetings

Messages such as “Dear User” can be suspicious when a legitimate service normally knows the customer’s identity.

A generic greeting alone does not prove phishing, but it can contribute to the overall assessment.

10.7 Requests for Credentials or OTPs

Requests for passwords, one-time passwords, authentication codes, or other sensitive information should be treated with caution.

Legitimate organizations generally have established secure methods for authentication and account management.

11. How a Typical Phishing Attack Works

A simplified phishing attack can follow this process:

Attacker creates deceptive message

↓

Message impersonates trusted organization

↓

Victim receives message

↓

Urgency / fear / curiosity is created

↓
Victim clicks link or opens attachment

↓
Fake website / malicious content

↓
Credentials or sensitive information may be exposed

The exact behavior depends on the attack and should not be assumed from the email alone.

12. Employee Awareness Guidelines

DO

- Verify the sender's complete email address.
- Check suspicious links before opening them.
- Navigate to important services using a known official website or application.
- Verify unexpected payment requests using another communication channel.
- Report suspicious emails to IT/security.
- Treat unexpected attachments cautiously.
- Use multi-factor authentication where available.
- Ask the security team when uncertain.
- Confirm unusual requests involving passwords or financial information.

DON'T

- Do not click unexpected links.
 - Do not enter passwords after following suspicious email links.
 - Do not share OTPs or authentication codes.
 - Do not open unexpected attachments.
 - Do not reply to suspicious messages.
 - Do not allow urgency to override verification procedures.
 - Do not trust a message only because it uses a familiar company logo.
 - Do not assume HTTPS alone means a website is legitimate.
 - Do not download files simply because an email claims they are important.
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13. What To Do If You Clicked a Phishing Link

If a user accidentally clicks a suspicious link:

1. Stop interacting with the page.
2. Do not enter credentials or sensitive information.
3. Close the suspicious page.
4. Report the incident to IT/security.
5. If credentials were entered, change the password through the official website.
6. Review account activity for suspicious behavior.
7. Follow the organization's incident-response procedure.

If sensitive information was submitted, the incident should be reported promptly rather than hidden.

14. Header Analysis

Email headers can provide useful information about message routing and authentication.

Relevant fields may include:

- From
- Reply-To
- Return-Path
- Received
- Message-ID
- Authentication-Results
- SPF
- DKIM
- DMARC

For this project, the sanitized versions of Samples 02 and 03 did not include complete raw headers.

Therefore, authentication results and originating infrastructure were not fabricated.

For a real investigation, analysts should obtain the complete original message headers and analyze them with an appropriate header-analysis tool.

15. URL and Attachment Safety

Potentially malicious URLs and attachments were not intentionally opened during this assessment.

Suspicious URLs should be represented in defanged form when included in documentation, for example:

`http://example[.]com`

instead of:

`http://example.com`

This reduces the chance of accidental navigation.

Similarly, suspicious attachments should not be opened on a normal personal system. If deeper analysis is required, security analysts should use an isolated analysis environment according to organizational policy.

16. Limitations

This assessment has several limitations:

1. The first sample is an educational example rather than a captured mailbox message.
2. Samples 02 and 03 are sanitized descriptions of public educational examples.
3. Complete raw headers were not available for all samples.
4. Potentially malicious URLs and attachments were not intentionally opened.
5. Classification is based on the evidence available in the samples.
6. No claim is made that the public samples were originally collected by the author of this report.

These limitations are documented so that the report does not claim evidence that was not actually observed.

17. Recommendations for Organizations

Organizations should:

- Provide regular phishing-awareness training.
 - Enable strong email authentication controls.
 - Use email filtering and anti-phishing protections.
 - Provide an easy phishing-reporting mechanism.
 - Encourage users to report mistakes without fear of punishment.
 - Use multi-factor authentication.
 - Establish procedures for suspicious payment and credential requests.
 - Conduct controlled security-awareness exercises.
 - Monitor reported phishing messages and recurring attack patterns.
 - Maintain an incident-response procedure for suspected phishing attacks.
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18. Practical Employee Checklist

Before interacting with an unexpected email, ask:

STOP

S – Stop and do not react immediately.

T – Think about whether you expected the message.

O – Observe the sender, links, attachments, and wording.

P – Proceed only after independently verifying the request.

If anything appears suspicious, report the message to the appropriate security or IT team.

19. Conclusion

The assessment demonstrates how phishing attacks commonly combine impersonation, urgency, fear, suspicious links, unexpected attachments, and trusted-brand references.

The most important defensive behavior is to slow down and independently verify unexpected requests.

Users should never assume that an email is legitimate simply because it contains familiar branding or appears urgent.

Phishing awareness is therefore an important part of organizational security because a well-informed user can prevent an attack before it progresses to credential theft or other compromise.

20. References

1. Future Interns – Cyber Security Task 2 (2026)
2. SeanWrightSec – phishing-examples
3. rf-peixoto – phishing_pot
4. Google Admin Toolbox – Messageheader
5. Google Gmail Help – Trace an email with its full header

See `../references/sources.md` for the source details and URLs.